(iii) 'n Dun ring met massa dm, volume dV en dikte dR se digtheid is ρ. Gee 'n uitdrukking vir dm in terme van ρ.

The mass of a thin ring is dm, *its volume is* dV *and its thickness is* dR. *The ring has a density ρ. Give an expression for* dm *in terms of ρ.*

$\rho = \frac{dm}{dV}$ $\rho = \frac{dm}{2\pi R\,dR\,h}$

$dm = \rho\, 2\pi R\,dR\,h$

(2)

(iv) Met bostaande uitdrukking vir dm, herlei 'n uitdrukking vir die traagheidsmoment I vir die wiel in terme van R_1 en R_2.

With the expression for dm *above, derive an expression for the moment of inertia* I *for the whole wheel in terms of* R_1 *and* R_2.

$I = \int R^2\, dm$

$= \int_{R_1}^{R_2} R^2\, \rho 2\pi R\, dR\, h$

$= \rho 2\pi h \int_{R_1}^{R_2} R^2 (R)\, dR$

$= 2\pi h \rho \int_{R_1}^{R_2} R^3\, dR$

$= 2\pi h \rho \left[\frac{1}{4} R^4 \right]_{R_1}^{R_2}$

$= 2\pi h \rho \left[\frac{1}{4} R_2^4 - \frac{1}{4} R_1^4 \right]$

5

(5)

(v) Indien die totale massa van die wiel M is en sy digtheid is ρ, skryf 'n uitdrukking vir ρ in terme van M, R_1 en R_2.

The wheel has a total mass M *and density ρ. Write an expression for ρ in terms of* M, R_1 *and* R_2.

$\rho = \frac{m}{\pi (R_2^2 - R_1^2) h}$

(1)

(vi) Met $R_2^4 - R_1^4 = (R_2^2 - R_1^2)(R_2^2 + R_1^2)$, toon aan dat $I = \frac{1}{2} M (R_2^2 + R_1^2)$.

With $R_2^4 - R_1^4 = (R_2^2 - R_1^2)(R_2^2 + R_1^2)$, *show that* $I = \frac{1}{2} M (R_2^2 + R_1^2)$.

$I = 2\pi h \rho \left[\frac{1}{4} R_2^4 - \frac{1}{4} R_1^4 \right]$

$= \frac{1}{2} \pi h \frac{m}{\pi (R_2^2 - R_1^2) h} \left[(R_2^2 - R_1^2)(R_2^2 + R_1^2) \right]$

$= \frac{1}{2} \frac{m}{(R_2^2 - R_1^2)} (R_2^2 - R_1^2)(R_2^2 + R_1^2)$

$= \frac{1}{2} M (R_2^2 + R_1^2)$

3

(3)

Vraag / *Question* 3(e)(vii)(vervolg / *continue*) / ...